THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 4: **Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following has the maximum valency? 1

(a) carbon (2, 4) **(b)** fluorine (2, 7) **(c)** oxygen (2, 6) **(d)** sodium (2, 8, 1)

2. $^{14}_{6}\mathrm{C}$ is used in: 1

(a) carbon dating **(b)** cancer therapy **(c)** nuclear fuel **(d)** goitre treatment

3. The ion S^{2-} has the electronic configuration: 1

(a) 2, 8, 8 **(b)** 2, 8, 7 **(c)** 2, 8, 6 **(d)** 2, 8

4. An element X with 5 valence electrons has valency: 1

(a) 8 **(b)** 3 **(c)** 2 **(d)** 5

5. The mass number of an isotope that has 8 protons and 10 neutrons is: 1

(a) 18 **(b)** 10 **(c)** 8 **(d)** 2

6. The number of neutrons in an atom of $^{39}_{19}\mathrm{K}$ is: 1

(a) 19 **(b)** 39 **(c)** 58 **(d)** 20

7. The number of valence electrons in sulphur (2, 8, 6) is: 1

(a) 6 **(b)** 8 **(c)** 2 **(d)** 16

8. The atomic number of the isobar of $^{40}_{18}\mathrm{Ar}$ that has 20 protons is: 1

(a) 22 **(b)** 18 **(c)** 20 **(d)** 40

9. The ion Al^{3+} has the following number of electrons: 1

(a) 10 **(b)** 16 **(c)** 3 **(d)** 13

10. The element with a stable octet and in the third shell is: 1

(a) argon **(b)** helium **(c)** chlorine **(d)** neon

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. An ion X^{-} has 18 electrons. Find the atomic number of X and the number of electrons in its neutral atom. 2

12. Why do elements with 8 electrons in the outer shell not combine easily? 2

13. The atomic number of an element is 13 and the mass number is 27. Find the number of electrons, protons and neutrons. 2

14. What is the difference between mass number and atomic mass? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how the neutral nature of the atom and the charge of the nucleus together show that the number of electrons equals the number of protons. 3

16. Explain with the example of Na and Cl how atoms attain stable configurations. 3

17. Differentiate between atomic number, mass number and atomic mass in three points. 3

OR

Explain why the atomic mass of chlorine is 35.5 u and not a whole number. 3

18. Describe how scientists know that the atom is mostly empty space. 3

OR

An ion of an element has 18 electrons and a charge of 2+. Identify the element and write the number of protons. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Compare the models of Thomson, Rutherford and Bohr in a table of five points: structure, charge, electrons, stability and drawbacks. 5

OR

An element has 4 shells, its second shell is full and the last shell has 1 electron. Find the configuration, atomic number, valency and name, and two properties. 5

20. Explain Rutherford's experiment with conclusions, and calculate the ratio of the atom's diameter (10^-10 m) to the nucleus diameter (10^-15 m). 5

OR

Explain the concept of nucleons and calculate the nuclear mass fraction of the mass of an atom contributed by the electrons for hydrogen (1 u vs 1/1836 u). 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND